MILITARY SPECIFICATION

PULP PROTECTOR, DENTAL, CALCIUM HYDROXIDE PASTE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a calcium hydroxide paste suitable as a dental pulp protector.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-C-186	Container, Packaging and Packing For Drugs, Chemicals and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6520

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

AMERICAN DENTAL ASSOCIATION

COUNCIL ON DENTAL THERAPEUTICS

Accepted Dental Therapeutics

(Application for copies should be addressed to the American Dental Association, 211 E. Chicago Avenue, Chicago, Illinois 60611).

3. REQUIREMENTS

3.1 Material. The paste shall be a suspension of calcium hydroxide adjusted to a smooth paste-like consistency with an aqueous solution of methylcellulose in distilled or deionized water. The paste shall contain no less than 47.5% nor more than 57.5% of calcium hydroxide by weight, when tested as specified in 4.4.1. All ingredients entering into the finished paste shall be of the Pharmacopeia of the United States of America (U.S.P.) quality.

3.2 Physical properties.

3.2.1 Appearance. The paste shall be a white, homogeneous paste which retains its shape and form on standing, when tested as specified in 4.4.2.

3.2.2 Setting time. The paste shall form a hardened adhesive coating with a setting time of three minutes maximum, when tested as specified in 4.4.3.

3.3 Performance. The paste shall be suitable for dental pulp capping, pulpotomy, and cavity lining under cement bases, silicate, and metallic fillings. The paste shall have the ability to stimulate the growth of secondary dentin and to promote odontoblastic regeneration as evidenced by listing of the paste as an acceptable calcium hydroxide preparation by the Council on Dental Therapeutics of the American Dental Association.

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3.4 Stability. The shelf life of the paste formulation shall be no less than eighteen months from date of delivery to the Government, when stored at controlled room temperature (59 - 86° Fahrenheit (F)).

3.5 Weight of contents. Each collapsible tube shall contain no less than 5.0 grams of paste.

3.6 Uniformity. The pulp protector paste shall have a smooth, creamy appearance, and shall be free from lumps and granules.

3.7 Toxicity. Prior to delivery, the manufacturer shall furnish a statement that the pulp protector paste, when clinically tested and used according to the directions supplied, has no toxic effect on a normal healthy person. This statement shall be sent directly to the Technical Services Branch (Code ATTS-2), Directorate of Medical Materiel, Defense Personnel Support Center, 2800 So. 20th Street, Philadelphia, PA 19101.

3.8 Compatibility. The immediate container (tube) and paste itself shall be compatible, and shall not interact physically or chemically with each other.

3.9 Cleanliness. The paste and the immediate container shall be free from all dirt, foreign matter, or other contaminants, both before and after filling.

3.10 Delivery schedule. No more than 2 months shall have elapsed from the date of manufacture of the paste to the date of delivery to the Government.

3.11 Instructions. Adequate and accurate instructions for use shall accompany each unit package. In addition, 2 copies of the instructions shall be forwarded to the contracting officer for information purposes.

3.12 Lot or control number. Each tube shall be marked with a lot or control number.

3.13 Workmanship. Workmanship shall be first class throughout. The product and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Lot. A lot or control is that single, uniform, and homogeneous quantity of the product subjected to the same manufacturing operation and filled into final containers.

4.2 Quality conformance inspection.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level S-2, AQL 1.5% defective for major and 4.0% defective for minor.

4.2.2 Sampling for tests. Sampling for tests shall be conducted on a sufficient quantity of paste from each lot or control number. Sampling for weight of contents shall be a minimum of 6 tubes (30 grams) taken at random from each lot or control number.

4.3 Examination. The paste shall be examined for defects including but not limited to those listed in table I.

Table I. Classification of Defects

Categories	Defects
Major:	
101	Paste not uniform in appearance and color. 1/
102	Paste not free from caked or set material. 1/
103	Paste and container not free from visible foreign matter. 1/

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Table I. Classification of Defects (cont'd)

Categories	Defects
Major:	
104	Immediate container torn or broken.
105	Immediate container shows leakage.
Minor:	
201	Immediate container not of classification specified
202	Packaging, packing, and marking not as specified.
203	Instructions missing or not as specified.

1/ In process examination may be used for this defect.

4.4 Tests.

4.4.1 Calcium hydroxide assay. The following method shall be f ignite a sample of 1.5 grams in a porcelain crucible, cool the ashe transfer by suitable means to a beaker, and dissolve by the gradual of 30 milliters (ml.) of diluted hydrochloric acid. When solution complete, transfer the solution to a 250 ml. volumetric flask, rins beaker thoroughly, adding the rinsings to the flask, dilute with wa volume, and mix. Pipet 50 ml. of the solution into a suitable cont add 100 ml. of water, 15 ml. of 1 Normal(1N) sodium hydroxide solut 40 milligrams (mg.) of murexide indicator preparation (a homogeneou of 400 mg. of murexide with 40 grams of powdered potassium sulfate) of Naphthol Green indicator solution (500 mg. of naphthol green B i water to make 1000 ml.), and titrate with 0.05 Molar (0.05 M) disod ethylenediaminetetraacetate until the solution is deep blue in colo ml. of 0.05 M disodium ethylenediaminetetraacetate is equivalent to 3.705 mg. of calcium hydroxide.

4.4.2 Appearance. The paste shall be expressed from the tube a ± 2.0° Centigrade (C.) (73.4 ± 3.6° F.) and tested for appearance characteristics specified in 3.2.1.

4.4.3 Setting time. The paste shall be expressed from the tube 23.0 ± 2.0° C. by spreading lightly on glass and air dried for one minute. The time interval from extrusion of the paste to formatio adhesive coating shall be recorded as the setting time. The setti time shall be recorded to the nearest half a minute and shall be t average of three determinations.

4.4.4 Weight. A minimum of 6 tubes shall be weighed before an filling or emptying of the tubes. The net weight shall be determi of the difference between the empty and filled tube.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One tube containing five grams of pulp protector, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Tube. Each tube shall conform to Group B, Class 2, type A, Closure E with an integral spreader as specified in PPP-C-186.

5.1.2.2 Unit package. Each tube shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure of the box shall be secured to prevent spilling of contents.

5.1.2.3 Intermediate package. Twelve unit packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure of the box shall be secured to prevent spilling of contents.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A. One hundred forty four intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. The box shall conform to the special requirements as specified in PPP-B-636. The box shall include liner, top and bottom pads. When intermediate boxes are stacked in two or more tiers, a pad shall be interposed between each tier. Liners and pads shall be fabricated of class domestic, double faced corrugated fiberboard having a minimum bursting strength of 200 pounds. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. One hundred forty four intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. The box shall conform to the special requirements as specified in PPP-B-636. The box shall include liners and pads as specified in 5.2.1. Closure of the box shall conform to method II as specified in PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Tube. Each tube shall be marked in accordance with commercial practice and shall include the following:

(a) Name of contents.
(b) Total quantity of contents - "5 grams".
(c) Lot or control number (may be embossed on heat-treated seal with or without color).
(d) Name and address of the manufacturer or contractor.

5.3.2 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable box is fabricated of water-resistant material. Contract or purchase order number, and name of contractor shall be shown. Lot (control) number and the following legend shall be shown: "STORE AT CONTROLLED ROOM TEMPERATURE (59 - 86° F.)".

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the following legend shall be shown: "STORE AT CONTROLLED ROOM TEMPERATURE (59 - 86° F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Stock No.	Item Identification
6520-00-817-2645	PULP PROTECTOR, DENTAL, Calcium Hydroxide Paste

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6.3 Supersession data. This specification includes the requirements of Purchase Description No. 2, for FSN 6520-817-2645, dated 21 April 1969.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1380

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-P-37088

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

YES NO (If "Yes", in what way?)

4 REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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